IMEN574 / IDSC574 · PROGRAMMING FOR AX

# 21

# Generative Models

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# Where We Are

01 **Programming with AI**

Lectures 2–4

02 **Python and Classical Machine Learning**

Lectures 5–13

03 **Deep Learning and Foundation Models**

Lectures 14–22

# Outline: Generative Models

01 **Sampling Distributions**

From one answer to many

02 **Denoising Models**

Learn a recovery field

03 **Flow and Editing**

Control a generation path

# 01 · Sampling Distributions


01 **Sampling Distributions**
From one answer to many

02 **Denoising Models**
Learn a recovery field

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# Why Model a Distribution

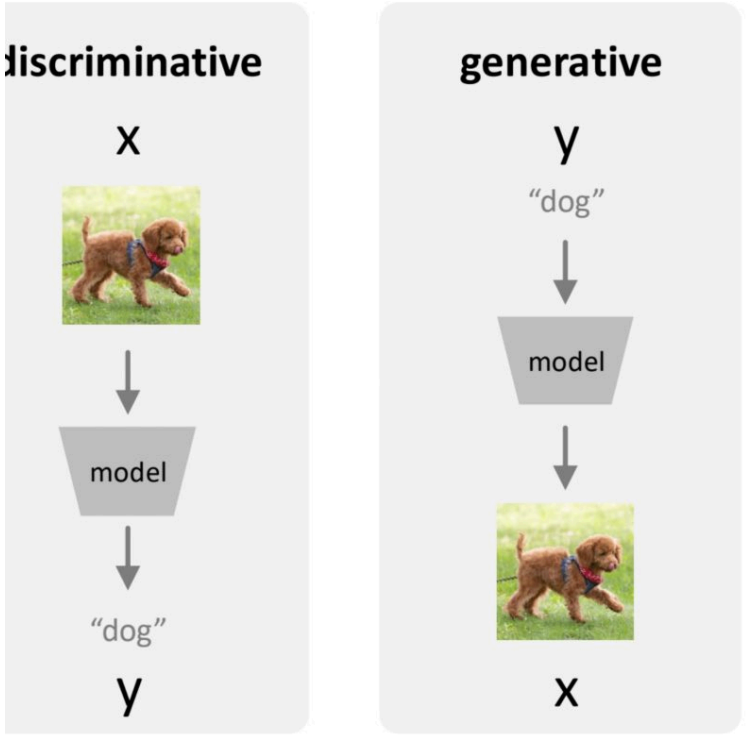
One condition can admit many valid outputs




Several outputs can satisfy the same condition

# Discriminative and Generative Tasks

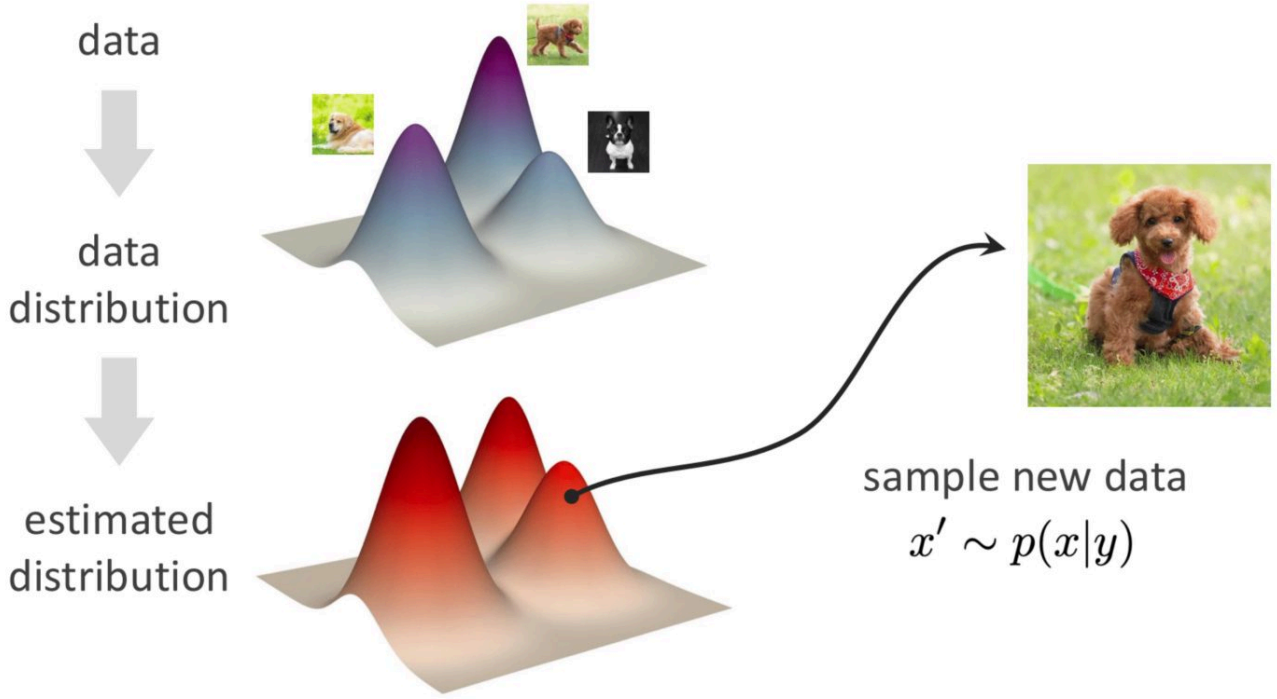
Prediction reverses direction when data become the output




The generated object is the data itself

# Data and Model Distributions

Sampling uses an approximation learned from observed data

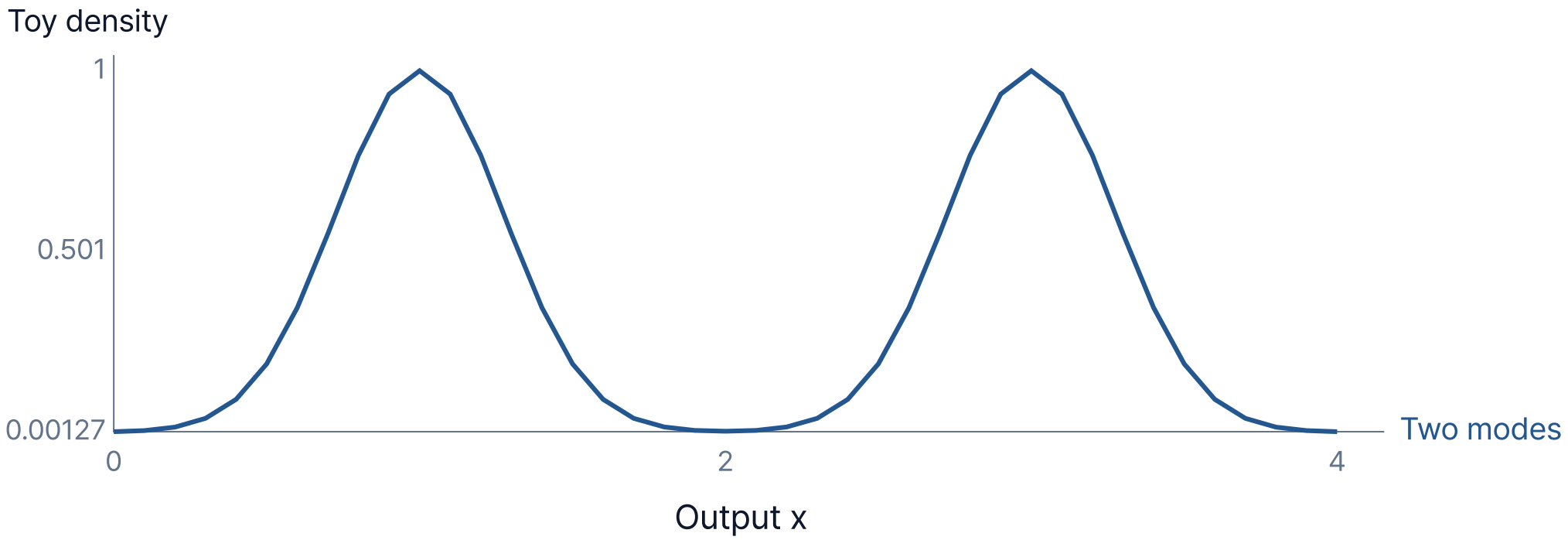



A learned density approximates observed variation

https://arxiv.org/abs/2006.11239

# Multimodal Targets

An average can fall between valid modes




The mean can lie between plausible modes

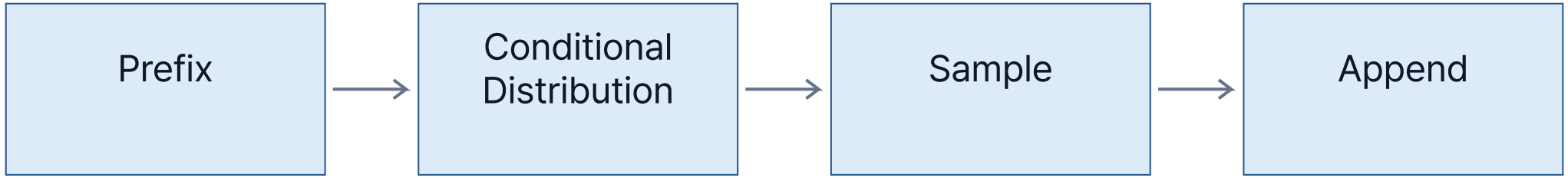
# Sampling and Randomness

A seed fixes draws under a fixed implementation

```
rng = np.random.default_rng(7)
samples = rng.normal(size=5)
assert samples.shape == (5,)
```

# Autoregressive Generation

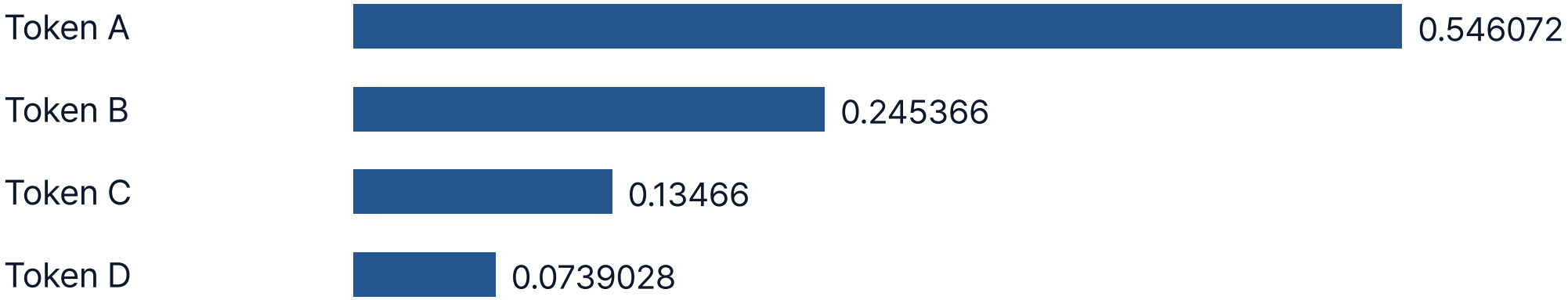
A joint distribution can be sampled through conditionals




A sampled prefix conditions the next draw

# Sampling Temperature

Rescaling logits changes probability concentration




Probability at temperature one

Temperature changes how strongly large logits dominate

# Demo: Sampling Temperature

Change temperature and compare the candidate probabilities

Temperature 1

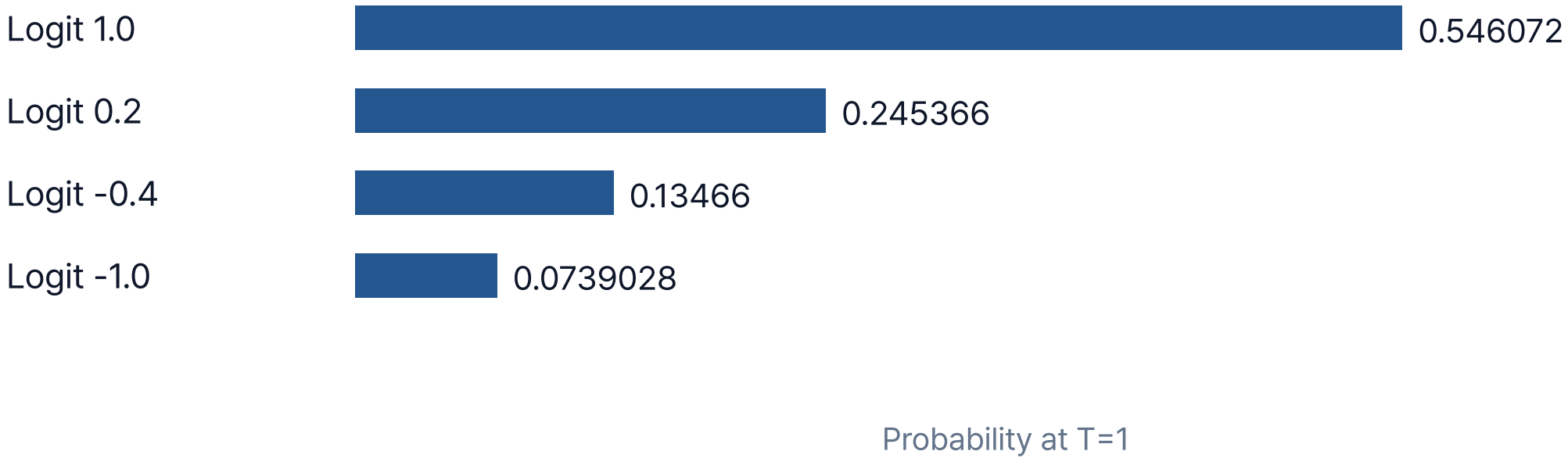



softmax([1, 0.2, −0.4, −1] / T) | entropy 1.138 nats

Higher temperature spreads probability; no text model runs in this demo

# Answer: Logit Probabilities

Softmax normalizes all four candidates




All four probabilities sum to one

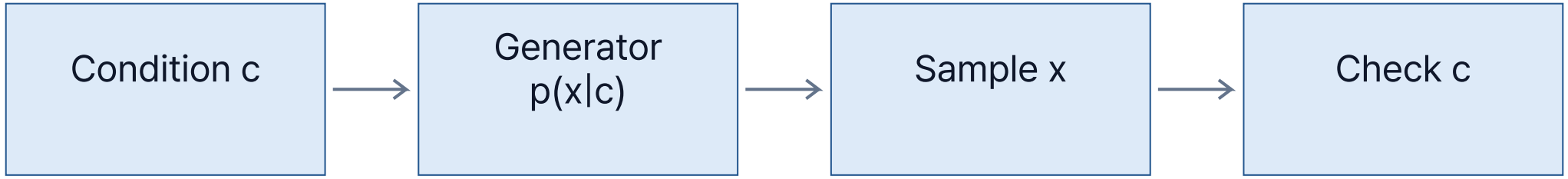
# Caveat: Diversity and Correctness

A wider sample distribution can include more invalid outputs

| Control | Changes | Does not certify |
| --- | --- | --- |
| Temperature | Concentration | Correctness |
| Seed | Random draw | Quality |
| More samples | Coverage | Best sample validity |

# Conditional Generation

A condition changes the distribution being sampled




The generated sample must still satisfy its condition

# 02 · Denoising Models

01 **Sampling Distributions**

From one answer to many

02 **Denoising Models**

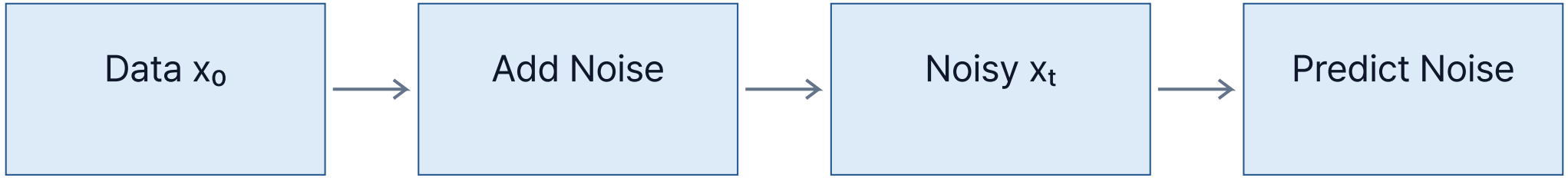
Learn a recovery field

03 **Flow and Editing**

Control a generation path

# Why Add Noise

A controlled corruption process creates a supervised recovery task




Known corruption supplies a training target

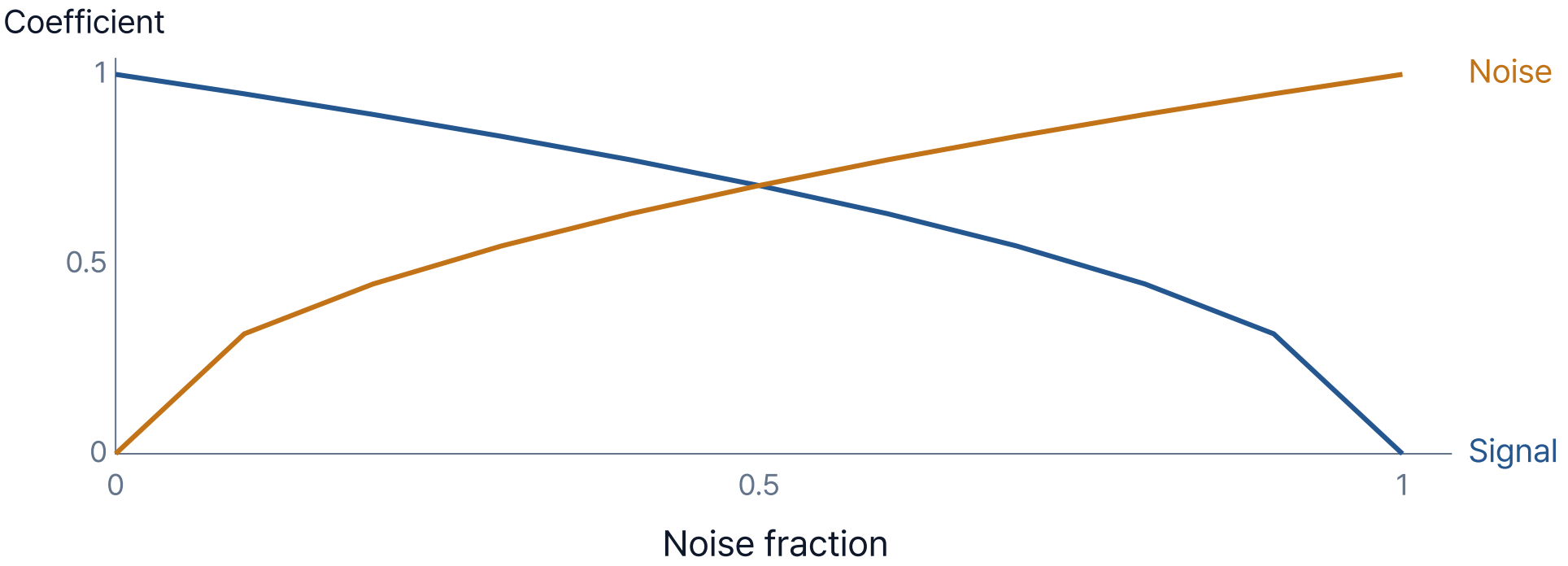
# Forward Noising

Signal strength shrinks while random noise grows

$$x_t = \sqrt{\bar{\alpha}_t}\, x_0 + \sqrt{(1 - \bar{\alpha}_t)}\, \varepsilon$$

# Signal and Noise Weights

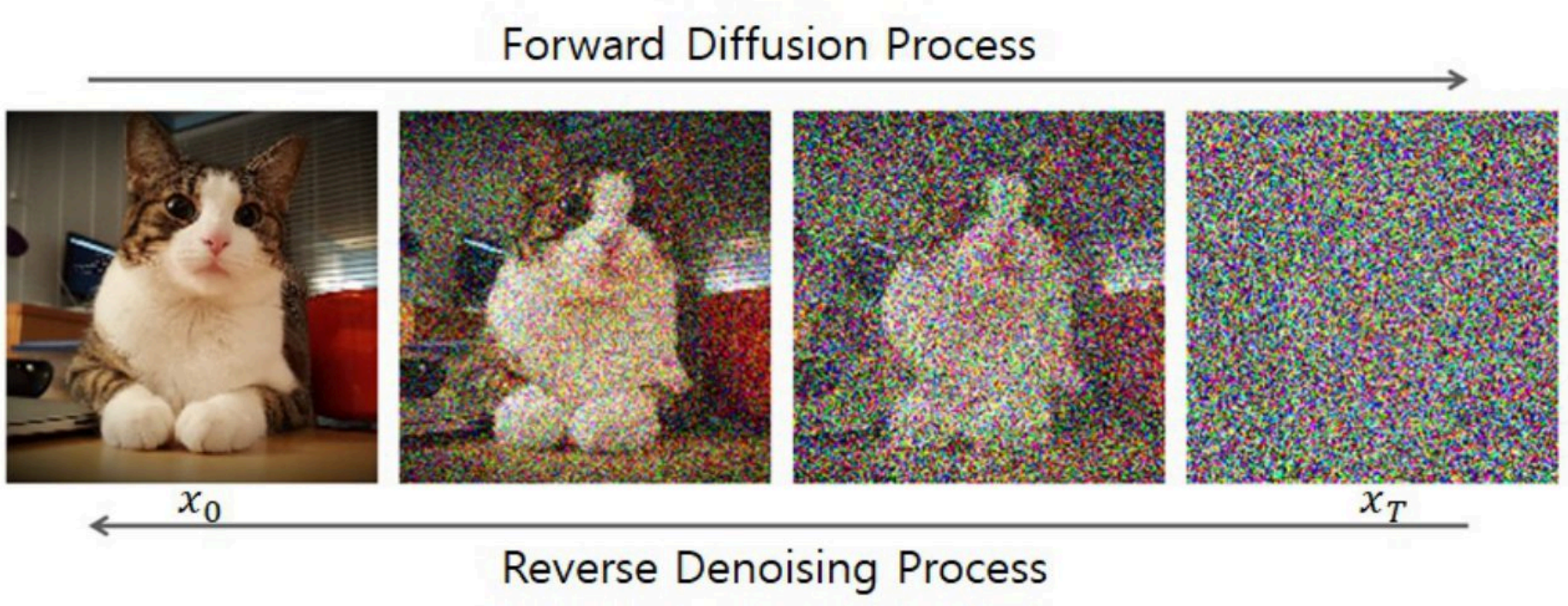
The schedule changes the mixture over time




The signal contribution fades as noise increases

# Denoising Diffusion

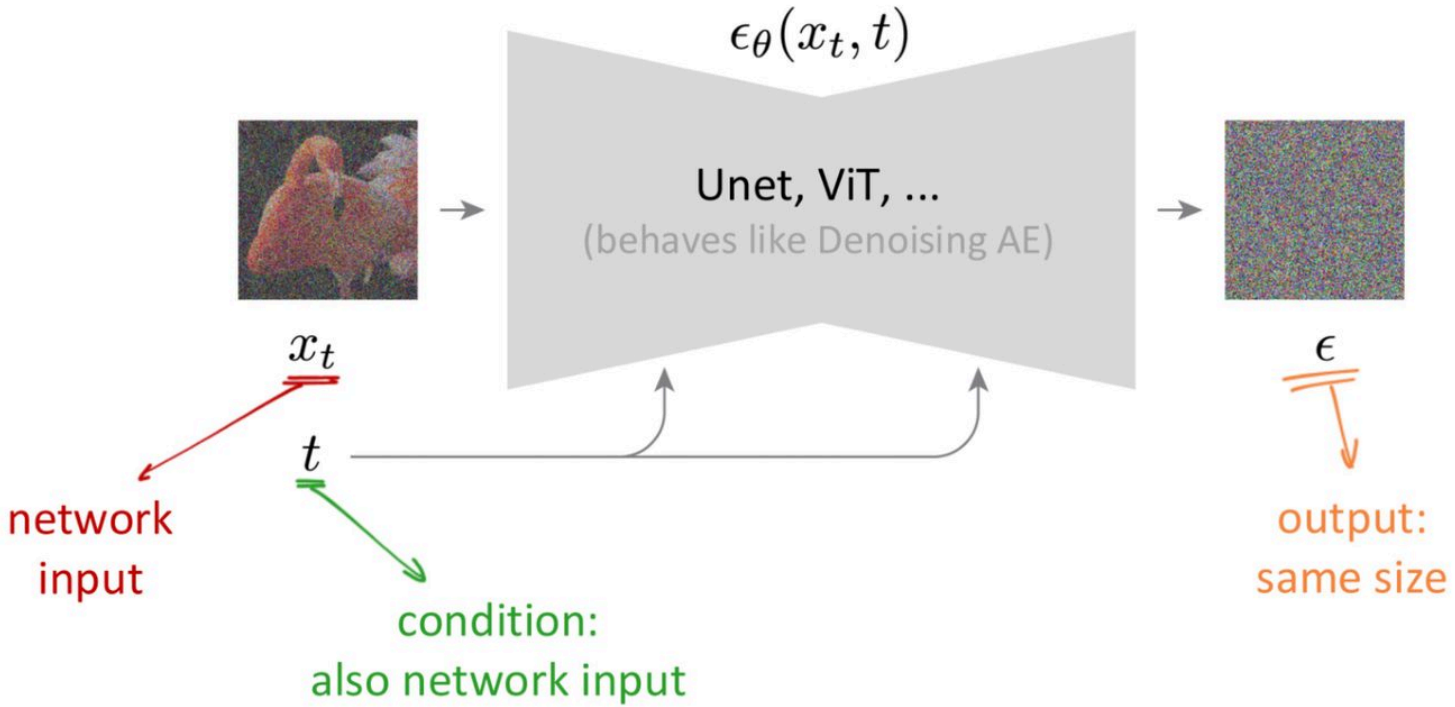
Generation reverses a gradual corruption process




The reverse path gradually recovers structure

# Noise Prediction Network

A noisy state and timestep enter the denoiser




The timestep tells the denoiser the corruption level

# Noise Prediction Loss

The observed noise provides a regression target

$$L = E[||\varepsilon - \varepsilon_\theta(x_t, t, c)||^2]$$

# A Noising Calculation

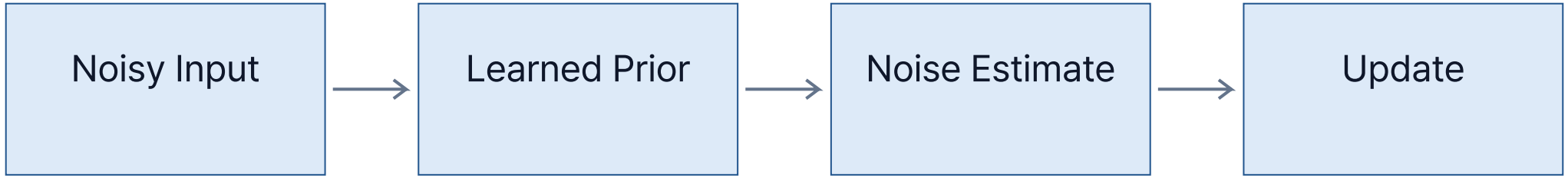
The same clean value can yield different noisy observations

| | $x_0$ | signal | noise | $\varepsilon$ | $x_t$ |
|---|---|---|---|---|---|
| Example | 2 | 0.6 | 0.8 | 0.5 | 1.6 |

Here $x_t = 0.6 \times 2 + 0.8 \times 0.5$

# Denoising Is Conditional

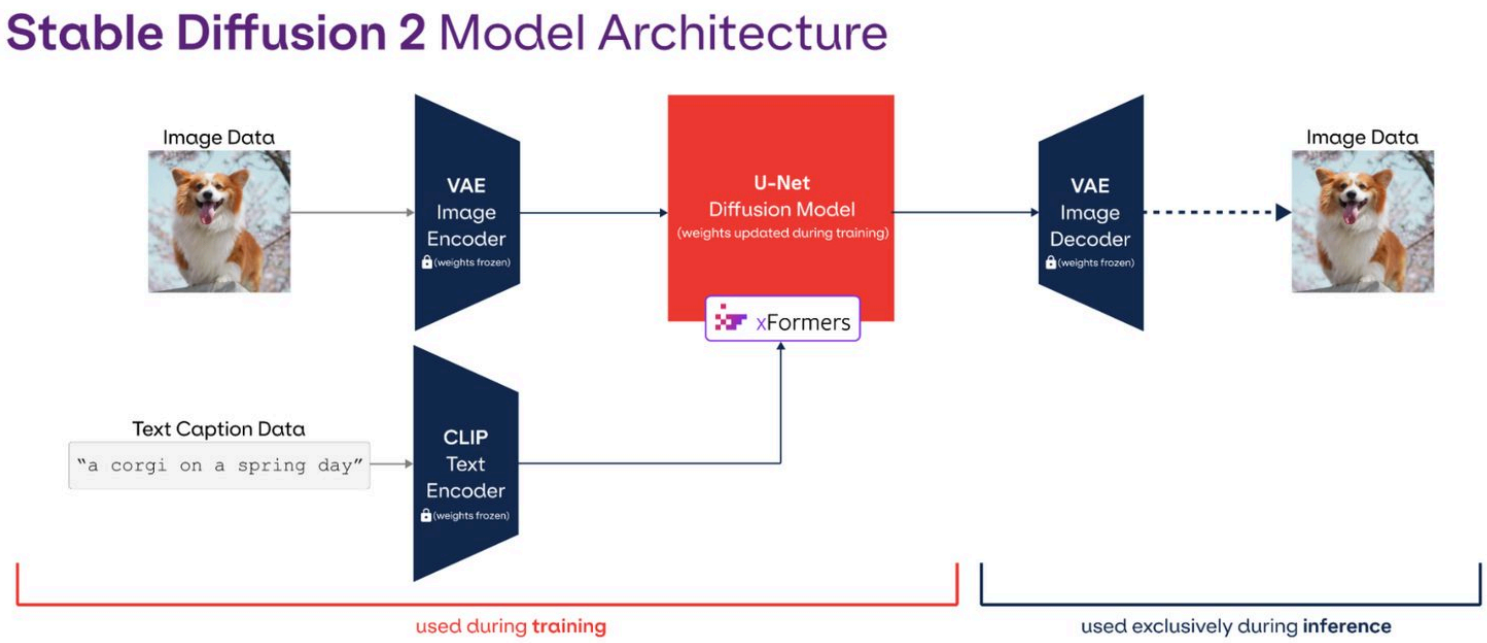
Noise recovery depends on the learned data prior




A learned prior guides the recovery step

# Latent Diffusion

A compact representation reduces the generation workspace

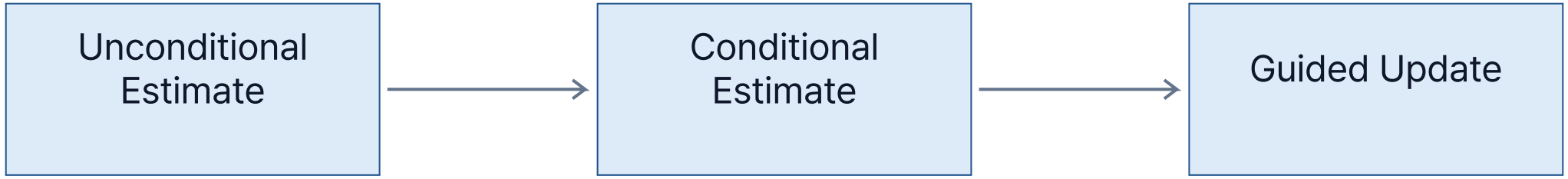



The denoiser works in a compressed image representation

https://arxiv.org/abs/2112.10752

# Conditional Guidance

Guidance combines conditional and unconditional predictions




A stronger guidance scale changes the sampling direction

# Caveat: Recovery Ambiguity

Severe corruption can hide multiple possible clean inputs

| | | | |
|---|---|---|---|
| Clean A | 1 | 0 | 1 |
| Clean B | 0 | 1 | 0 |
| Noisy observation | 0.5 | 0.5 | 0.5 |

A plausible reconstruction need not be the original

# 03 · Flow and Editing

01 Sampling Distributions

From one answer to many

02 Denoising Models

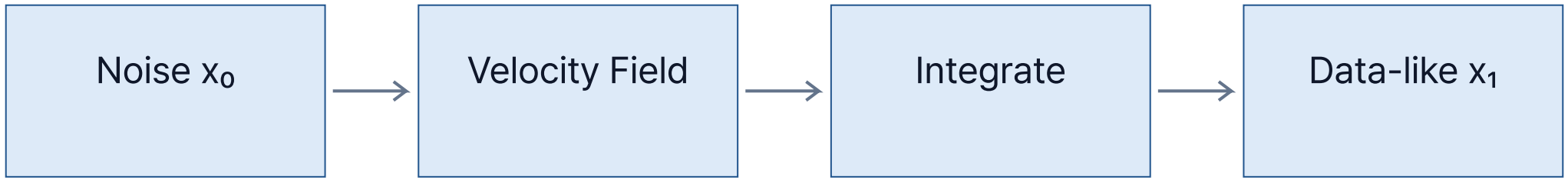
Learn a recovery field

03 **Flow and Editing**

Control a generation path

# Why Predict a Flow

A velocity field can transport noise toward data




A learned velocity defines a transport path

# Linear Interpolation Path

A simple path joins a noise sample and data sample

$$x_t = (1 - t)x_{noise} + tx_{data}$$

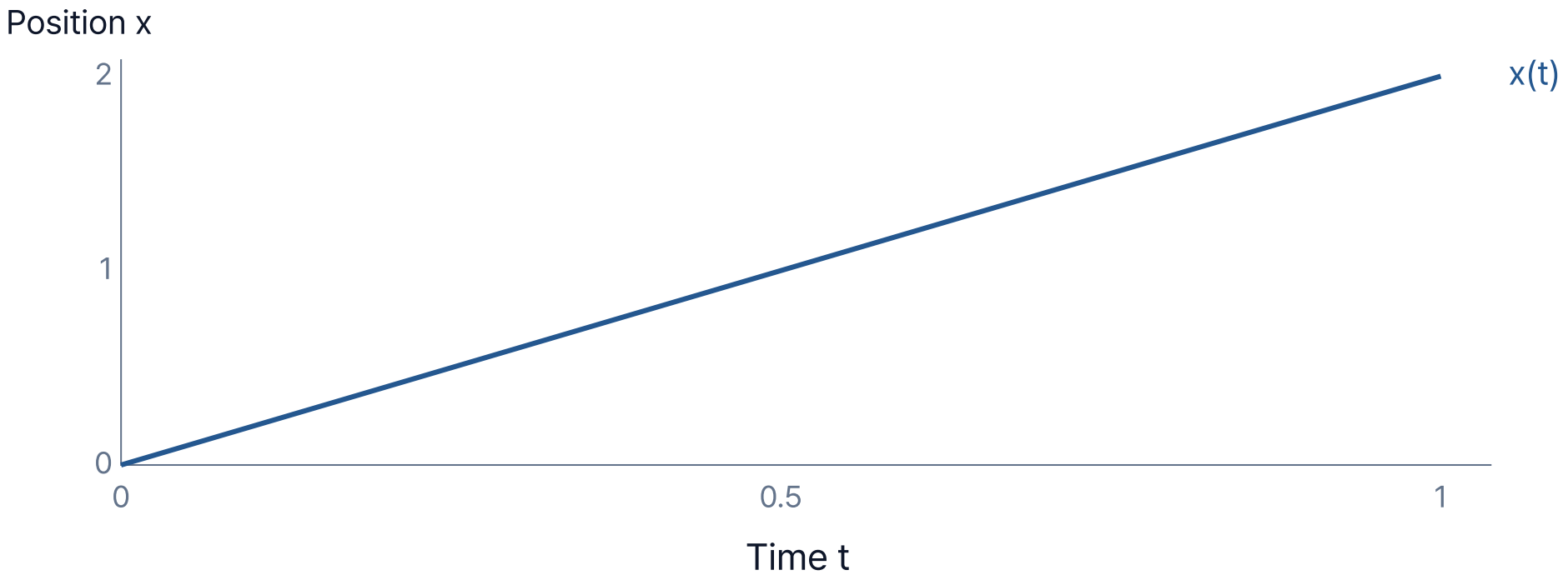
# Velocity Target

The linear path has a constant target velocity

```
v_target = x_data - x_noise
```

# Euler Integration

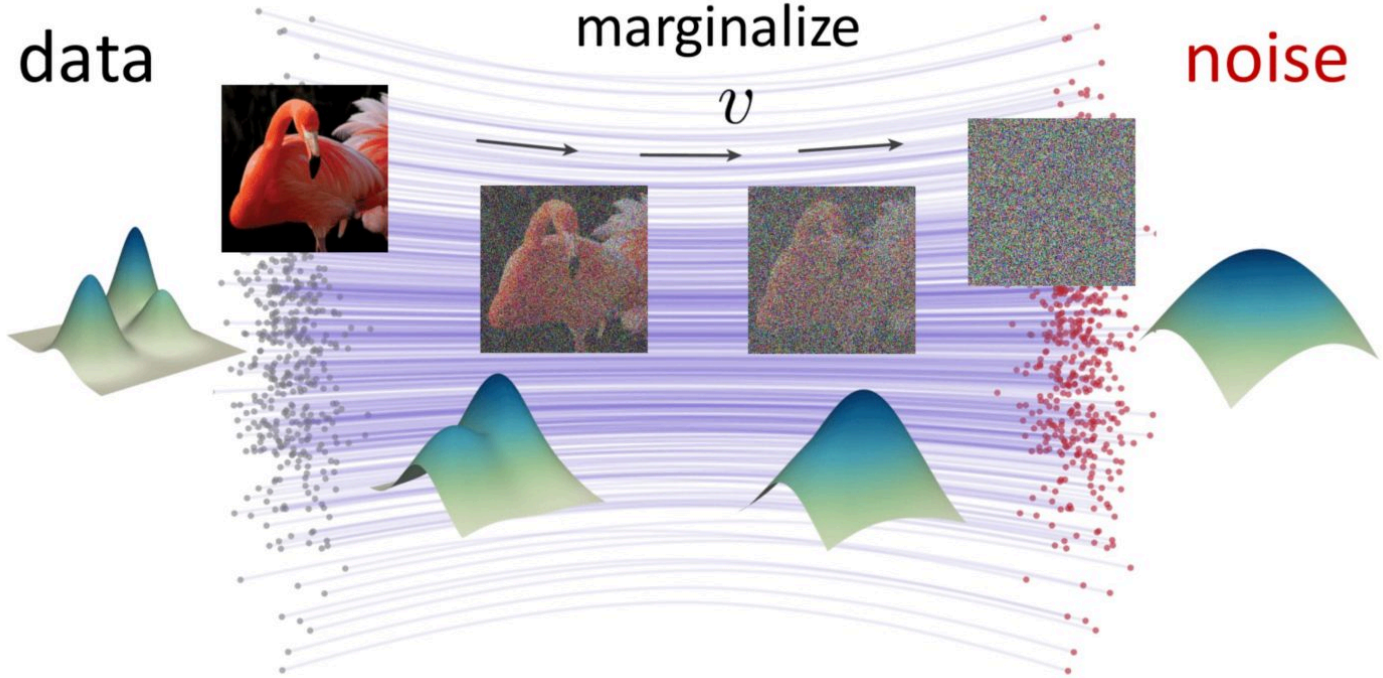
A numerical solver turns velocity into successive positions




Position advances by step size times velocity

# Flow Matching Illustration

The learned vector field connects distributions




A velocity field moves a distribution over time

https://arxiv.org/abs/2210.02747

# Distillation Tradeoff

A shorter generation path shifts work into training

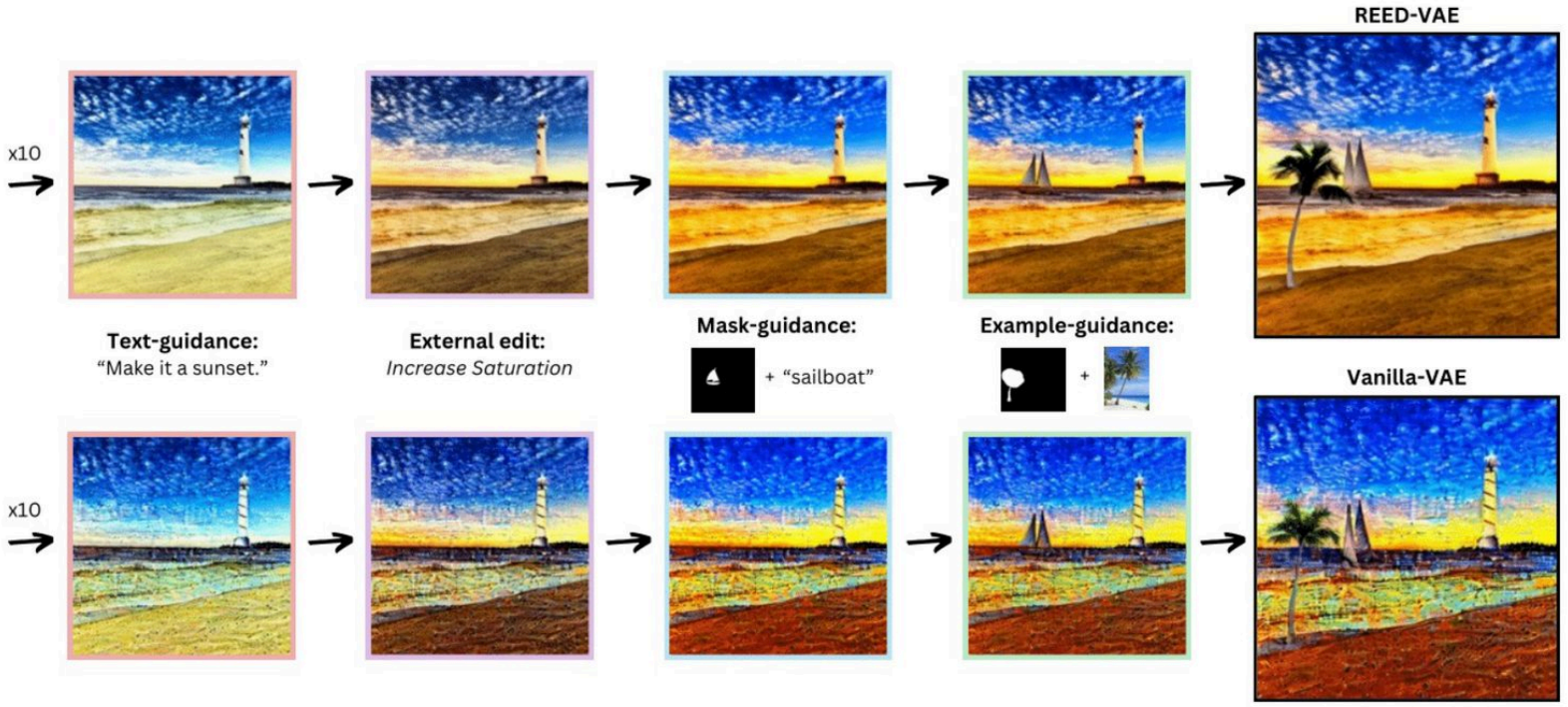



Illustrative model evaluations

Fewer evaluation steps do not remove the quality check

# Iterative Image Editing

A previous image can condition a sequence of edits




Each edit must preserve the unconstrained content

# Unified Multimodal Generation

A shared model can couple understanding and image production

| Capability | Input | Output |
| --- | --- | --- |
| Understanding | Image + question | Text |
| Generation | Text or image context | Image |
| Editing | Image + instruction | Revised image |

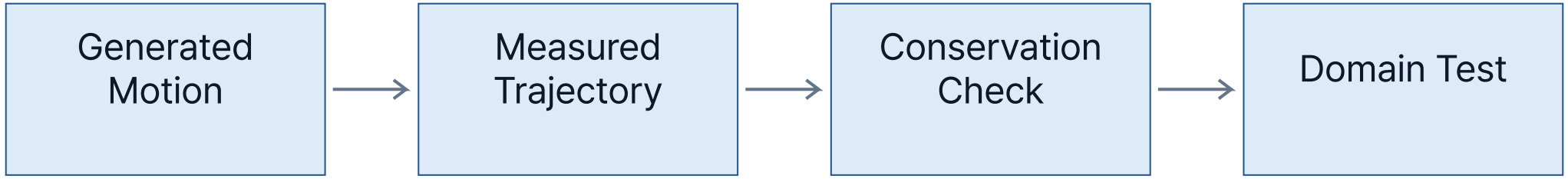
# Generative Evaluation

Assess validity, diversity and data resemblance separately

| Question | Test |
| --- | --- |
| Condition satisfied? | Targeted checks |
| Samples vary? | Repeated seeds |
| Data-like outputs? | Held-out comparison |

# Caveat: Simulation Claims

Plausible video alone does not establish physical accuracy




Physical claims require measurements beyond appearance

# Takeaway: Generative Models

01 A generated sample is drawn from a learned approximation

02 Training targets and sampling rules serve different roles

03 Condition fidelity and physical validity need explicit checks